

Math 714 — Assignment 0

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Instructions

This document contains my solutions and computational work for Assignment 0 of Math 714. All problem statements are reproduced below for clarity, followed by my solutions.

Problem 1

Consider the recursive sequence

$$x_{k+1} = ax_k^2 + bx_k, \quad k = 0, 1, 2, \dots$$

- (a) If the sequence converges, to what value or values does it converge?
- (b) Suppose that x_0 is arbitrarily close to the value or values above. For what parameters (a, b) does the sequence x_k converge to that value?

Solution

Problem 2

The Chebyshev polynomials $T_k(x)$ satisfy the recurrence

$$T_k(x) = 2x T_{k-1}(x) - T_{k-2}(x), \quad T_0(x) = 1, \quad T_1(x) = x.$$

Evaluate and plot $T_5(x)$ at 101 evenly spaced points in $[-1, 1]$. Draw a 2D surface plot of $T_3(x)T_5(y)$ on a 101×101 grid over $(x, y) \in [-1, 1]^2$.

Solution

Problem 3

The IEEE double-precision standard guarantees that for any operation $*$, the floating point operation $\circledast$ satisfies

$$x \circledast y = (1 + \delta)(x * y), \quad |\delta| < \epsilon.$$

- (a) Compute the minimum and maximum possible values of

$$S = \frac{3}{4 + 2}$$

when evaluated using floating point arithmetic as

$$\tilde{S} = 3 \oslash (4 \oplus 2).$$

- (b) Show that if $O(\epsilon^2)$ terms are neglected, then

$$|S - \tilde{S}| < \lambda \epsilon,$$

and determine the constant λ .

Solution

Problem 4

For an invertible matrix A , define the condition number

$$\kappa(A) = \|A\| \|A^{-1}\|.$$

Assume the matrix norm is induced by the Euclidean vector norm.

- (a) Find 2×2 invertible matrices B and C such that $\kappa(B + C) < \kappa(B) + \kappa(C)$.
- (b) Find 2×2 invertible matrices B and C such that $\kappa(B + C) > \kappa(B) + \kappa(C)$.
- (c) Suppose A is symmetric and invertible. Find $\kappa(A^2)$ in terms of $\kappa(A)$.
- (d) Does the result from part (c) hold if A is not symmetric? Either prove the result or give a counterexample.
- (e) For invertible matrices B and C , prove that $\kappa(BC) \leq \kappa(B)\kappa(C)$. Give examples where equality holds and where the inequality is strict.

Solution

Problem 5

The gamma function is defined by

$$\Gamma(x) = \int_0^{\infty} t^{x-1} e^{-t} dt.$$

It satisfies $(n-1)! = \Gamma(n)$ for integers n . Also,

$$\Gamma\left(\frac{3}{2}\right) = \frac{1}{2}\sqrt{\pi}.$$

- (a) For $k = 1, \dots, 5$, compute polynomials $p_k(x)$ of degree k matching $\Gamma(x)$ at $x = 1, 2, \dots, k+1$. Evaluate the error $|p_k(\frac{3}{2}) - \Gamma(\frac{3}{2})|$.
- (b) For $k = 1, \dots, 5$, compute polynomials $q_k(x)$ of degree k matching $\log \Gamma(x)$ at $x = 1, 2, \dots, k+1$. Evaluate the error $|\exp(q_k(\frac{3}{2})) - \Gamma(\frac{3}{2})|$.
- (c) Examine the asymptotic behavior of both errors as $k \rightarrow \infty$.

Solution

Problem 6

- (a) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be five-times differentiable. Expand Taylor series for $f(x-h)$ and $f(x+h)$ to derive coefficients α, β, γ such that

$$f''(x) = \frac{\alpha f(x-h) + \beta f(x) + \gamma f(x+h)}{h^2} + O(h).$$

- (b) Suppose f' can be evaluated exactly. Find coefficients a, b, c, r, s, t such that

$$\begin{aligned} f''(x) = & \frac{af(x-h) + bf(x) + cf(x+h)}{h^2} \\ & + \frac{rf'(x-h) + sf'(x) + tf'(x+h)}{h} + O(h^4). \end{aligned}$$

- (c) Consider $f(x) = e^{4 \sin x}$. Compute f' and f'' analytically. Write a program to test the formulas in (a) and (b) at $x = 1$ using $h = 2^{-k}$ for $k = 0, 1, \dots, 23$. Make log-log plots of the error E vs. h and fit $E = Ch^p$. Discuss whether the observed p matches the theoretical order.

Solution